Hydrogen Ground-State Hyperfine Splitting as a Canonical Activation Gap: From the Fermi Contact Interaction to the Schottky Heat-Capacity Anomaly

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Abstract

We give a self-contained, orthodox derivation of the hydrogen $1s$ hyperfine splitting starting from the magnetic dipole field (including its distributional contact term) and the resulting Fermi contact Hamiltonian. The leading-order gap ΔE_{hfs} is expressed in standard constants, recovering the well-known scaling $\Delta E_{\mathrm{hfs}} \propto \alpha^4 (m_e/m_p) m_e c^2$. We then treat the $(F=0)$–$(F=1)$ manifold as a two-level system with degeneracies $(g_0, g_1) = (1, 3)$ and derive the exact canonical heat capacity (Schottky anomaly), including its low-temperature asymptotic and an explicit derivation of the peak condition for the degeneracy-asymmetric case. Numerical evaluation yields $\Delta E_{\mathrm{hfs}} \approx 5.88\ \mu$eV from the leading term, to be compared with the experimental value 5.8743 μeV (fractional error $\approx$ 0.097%), with the remaining discrepancy attributable to known QED, recoil, and proton-structure corrections.

Keywords: hydrogen hyperfine structure; Fermi contact interaction; Schottky anomaly; two-level system; canonical ensemble; low-temperature thermodynamics; atomic spectroscopy.

1 Introduction

The hydrogen ground-state hyperfine splitting provides a concrete, experimentally calibrated example of a gapped two-level subsystem. Independently, the Schottky anomaly describes the characteristic heat-capacity peak arising from discrete-level occupation in the canonical ensemble. Both topics are standard, but they are often presented separately (atomic structure vs. thermal physics). The purpose of this manuscript is to present a single, fully explicit derivation chain that (i) derives the leading-order hyperfine gap from the Fermi contact interaction and (ii) propagates that gap into thermodynamic predictions for the two-level heat capacity, including an explicit derivation of the peak condition for the degeneracy-asymmetric case $(g_0, g_1) = (1, 3)$. Accordingly, the contribution is pedagogical and synthetic rather than one of new fundamental physics: it provides a closed analytic-to-thermodynamic derivation chain from the hydrogen $1s$ hyperfine gap to the exact degeneracy-asymmetric Schottky anomaly, with explicit millikelvin scales relevant to low-temperature spectroscopy and thermodynamics.

2 Definitions and standard identities

We use SI units. The fundamental constants are: c, $\hbar$, $h = 2\pi\hbar$, e, ε_0, $\mu_0 = 1/(\varepsilon_0 c^2)$, k_B, and the electron and proton masses (m_e, m_p).

The fine-structure constant is

$$\alpha \equiv \frac{e^2}{4\pi\varepsilon_0 \hbar c}. \tag{1}$$

The Bohr radius is

$$a_0 \equiv \frac{4\pi\varepsilon_0 \hbar^2}{m_e e^2}. \tag{2}$$

Using (1) (i.e. $e^2 = 4\pi\varepsilon_0 \hbar c\, \alpha$), one obtains the standard identity

$$a_0 = \frac{\hbar}{m_e c}\,\frac{1}{\alpha}. \tag{3}$$

The electron and proton magnetic moments are written using g-factors:

$$\boldsymbol{\mu}_e = -g_e \mu_B \frac{\mathbf{S}}{\hbar}, \qquad \boldsymbol{\mu}_p = +g_p \mu_N \frac{\mathbf{I}}{\hbar}, \tag{4}$$

where $\mathbf{S}$ is the electron spin operator and $\mathbf{I}$ is the proton spin operator, and

$$\mu_B \equiv \frac{e\hbar}{2m_e}, \qquad \mu_N \equiv \frac{e\hbar}{2m_p}. \tag{5}$$

For hydrogen $1s$, we use $g_e \simeq 2$ and $g_p \simeq 5.5856947$ (proton g-factor) for the leading-order estimate; the effect of using the free-electron value $g_e = 2.002319304\ldots$ is discussed in Section 5 [1].

The normalized $1s$ wavefunction is

$$\psi_{1s}(\mathbf{r}) = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}, \qquad |\psi_{1s}(0)|^2 = \frac{1}{\pi a_0^3}. \tag{6}$$

3 Magnetic dipole interaction and the Fermi contact term

3.1 Dipole field and contact contribution

The magnetic field of a point dipole $\boldsymbol{\mu}$ at the origin contains a distributional (contact) term. In SI units one may write [2]

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi r^3}\Big[3\hat{\mathbf{r}}(\boldsymbol{\mu}\cdot\hat{\mathbf{r}}) - \boldsymbol{\mu}\Big] + \frac{2\mu_0}{3}\,\boldsymbol{\mu}\,\delta^{(3)}(\mathbf{r}). \tag{7}$$

For an s-state, the isotropic contact term is the relevant contribution to the hyperfine splitting.

3.2 Fermi contact Hamiltonian and consistent sign convention

To avoid sign ambiguity, we compute the interaction as the energy of the electron magnetic moment in the proton's magnetic field:

$$H_{\text{contact}} = -\boldsymbol{\mu}_e \cdot \mathbf{B}_p^{(\text{contact})}. \tag{8}$$

Using the proton contact field from (7),

$$\mathbf{B}_p^{(\text{contact})} = \frac{2\mu_0}{3}\,\boldsymbol{\mu}_p\,\delta^{(3)}(\mathbf{r}), \tag{9}$$

we obtain

$$H_{\text{contact}} = -\frac{2\mu_0}{3}\,(\boldsymbol{\mu}_e\cdot\boldsymbol{\mu}_p)\,\delta^{(3)}(\mathbf{r}). \tag{10}$$

The electron and proton magnetic moments are

$$\boldsymbol{\mu}_e = -g_e \mu_B \frac{\mathbf{S}}{\hbar}, \qquad \boldsymbol{\mu}_p = +g_p \mu_N \frac{\mathbf{I}}{\hbar}, \tag{11}$$

so that

$$\boldsymbol{\mu}_e \cdot \boldsymbol{\mu}_p = -\, g_e g_p\, \mu_B \mu_N\, \frac{\mathbf{S}\cdot\mathbf{I}}{\hbar^2}. \tag{12}$$

Substituting into (10) gives

$$H_{\text{contact}} = +\frac{2\mu_0}{3}\, g_e g_p\, \mu_B \mu_N\, \frac{\mathbf{S}\cdot\mathbf{I}}{\hbar^2}\, \delta^{(3)}(\mathbf{r}). \tag{13}$$

Taking the expectation value in the $1s$ spatial state yields

$$\langle H_{\text{contact}}\rangle_{1s} = A\,\mathbf{S}\cdot\mathbf{I}, \qquad A \equiv \frac{2\mu_0}{3}\, g_e g_p\, \mu_B \mu_N\, \frac{|\psi_{1s}(0)|^2}{\hbar^2}, \tag{14}$$

with $A > 0$.

The energy eigenstates are labeled by total spin $\mathbf{F} = \mathbf{S} + \mathbf{I}$ with $S = I = 1/2$ and $F \in \{0, 1\}$. Using

$$\mathbf{S}\cdot\mathbf{I} = \frac{1}{2}\Big(F(F+1) - S(S+1) - I(I+1)\Big)\hbar^2, \tag{15}$$

and $S(S+1) = I(I+1) = 3/4$, one finds

$$\langle \mathbf{S}\cdot\mathbf{I}\rangle_{F=1} = +\frac{1}{4}\hbar^2, \qquad \langle \mathbf{S}\cdot\mathbf{I}\rangle_{F=0} = -\frac{3}{4}\hbar^2. \tag{16}$$

Define

$$\mathcal{A} \equiv A\hbar^2 = \frac{2\mu_0}{3}\, g_e g_p\, \mu_B \mu_N\, |\psi_{1s}(0)|^2. \tag{17}$$

Then

$$\Delta E_{\text{hfs}} \equiv E_{F=1} - E_{F=0} = \mathcal{A}\left(\frac{1}{4} - \left(-\frac{3}{4}\right)\right) = \mathcal{A}. \tag{18}$$

Since $\mathcal{A} > 0$, the singlet lies below the triplet, and therefore

$$E_{F=0} < E_{F=1}, \qquad \Delta E_{\text{hfs}} > 0.$$

Substituting $|\psi_{1s}(0)|^2 = 1/(\pi a_0^3)$ and the definitions of μ_B, μ_N yields the standard leading-order Fermi-contact formula

$$\Delta E_{\text{hfs}} = \frac{2\mu_0}{3}\, g_e g_p\, \mu_B \mu_N\, \frac{1}{\pi a_0^3}. \tag{19}$$

For a closely related pedagogical derivation and sign-consistent discussion, see Griffiths [3].

4 Closed-form expression in α

We eliminate μ_0, μ_B, μ_N, a_0 in favor of (α, m_e, m_p, c).

Use $\mu_0 = 1/(\varepsilon_0 c^2)$ and (5):

$$\mu_0\, \mu_B \mu_N = \frac{1}{\varepsilon_0 c^2}\left(\frac{e\hbar}{2m_e}\right)\left(\frac{e\hbar}{2m_p}\right) = \frac{e^2\hbar^2}{4\varepsilon_0 c^2\, m_e m_p}. \tag{20}$$

Insert into (19):

$$\Delta E_{\text{hfs}} = \frac{g_e g_p}{6\pi}\, \frac{e^2\hbar^2}{\varepsilon_0 c^2\, m_e m_p\, a_0^3}. \tag{21}$$

Now eliminate a_0 using (2):

$$a_0 = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2} \quad\Longrightarrow\quad \frac{1}{a_0^3} = \left(\frac{m_e e^2}{4\pi\varepsilon_0\hbar^2}\right)^3.$$

Table 1: Constants used in the leading-order numerical estimate of $\Delta E_{\rm hfs}^{(0)}$.

Quantity	Value	Source
α	$7.2973525643 \times 10^{-3}$	[1]
$m_e c^2$	510998.950 eV	[1]
m_e/m_p	$5.446170214 \times 10^{-4}$	[1]
g_e	2 (leading-order)	convention in this paper
g_p	5.5856947	[1]

Substitute:

$$\begin{aligned}\Delta E_{\rm hfs} &= \frac{g_e g_p}{6\pi}\,\frac{e^2\hbar^2}{\varepsilon_0 c^2\, m_e m_p}\left(\frac{m_e e^2}{4\pi\varepsilon_0\hbar^2}\right)^3 \\ &= \frac{g_e g_p}{6\pi}\,\frac{m_e^2}{m_p}\,\frac{e^8}{(4\pi)^3\,\varepsilon_0^4\,\hbar^4}\,\frac{1}{c^2}.\end{aligned} \tag{22}$$

Using $e^2/(4\pi\varepsilon_0) = \alpha\hbar c$, one has

$$\frac{e^8}{(4\pi)^3\varepsilon_0^4} = (4\pi)(\alpha\hbar c)^4,$$

so that

$$\begin{aligned}\Delta E_{\rm hfs} &= \frac{g_e g_p}{6\pi}\,\frac{m_e^2}{m_p}\,\frac{(4\pi)(\alpha\hbar c)^4}{\hbar^4 c^2} \\ &= \frac{4\pi}{6\pi}\,g_e g_p\,\alpha^4\left(\frac{m_e}{m_p}\right)m_e c^2 = \frac{2}{3}\,g_e g_p\,\alpha^4\left(\frac{m_e}{m_p}\right)m_e c^2.\end{aligned} \tag{23}$$

Thus the compact leading-order form is

$$\boxed{\Delta E_{\rm hfs} = \frac{2}{3}\,g_e g_p\,\alpha^4\left(\frac{m_e}{m_p}\right)m_e c^2.} \tag{24}$$

Literature note (prior derivations). Equation (24) is the standard leading-order Fermi-contact result ("Fermi energy") and is derived in many classic sources (e.g. Berestetskii–Lifshitz–Pitaevskii) as well as modern reviews. We reproduce the algebra here to keep the manuscript self-contained and to connect the gap to the thermodynamic analysis below [4, 5].

5 Numerical evaluation and the 21 cm line

Using CODATA constants and g_p [1], (24) gives a leading-order estimate

$$\Delta E_{\rm hfs}^{(0)} \approx 5.88\ \mu\text{eV}. \tag{25}$$

Experimentally, the hydrogen ground-state hyperfine transition frequency is

$$\nu_{\rm hfs} = 1420.40575177\ \text{MHz}, \tag{26}$$

corresponding to

$$\Delta E_{\rm hfs}^{(\rm exp)} = h\nu_{\rm hfs} \approx 5.8743\ \mu\text{eV}. \tag{27}$$

(See primary measurement references such as Hellwig et al. [6].)

Fractional deviation and input conventions. Define

$$\delta \equiv \frac{\Delta E_{\rm hfs}^{(0)} - \Delta E_{\rm hfs}^{(\rm exp)}}{\Delta E_{\rm hfs}^{(\rm exp)}}.$$

Using the leading-order contact formula (24) with the conventional approximation $g_e \simeq 2$ gives $\Delta E_{\rm hfs}^{(0)} \approx 5.88\ \mu$eV and hence

$$\delta \approx 9.7 \times 10^{-4} \approx 0.097\%.$$

If instead one inserts the free-electron value $g_e = 2.002319304\ldots$ into the same leading-order expression, $\Delta E_{\rm hfs}^{(0)}$ shifts by 0.116% and the residual changes at the $\sim 10^{-3}$ level. In precision theory one does not treat (24) as complete: radiative QED corrections, recoil effects, and proton-structure (Zemach-radius) contributions are required for ppm-level agreement [5, 4].

Define the gap temperature scale

$$T_\Delta \equiv \frac{\Delta E_{\rm hfs}^{(\rm exp)}}{k_B} \approx 68.17\ \text{mK}. \tag{28}$$

6 Thermodynamics: Schottky anomaly for the hyperfine doublet

6.1 Two-level system with degeneracy $(1, 3)$

The $1s$ hyperfine manifold has a singlet ground state ($F = 0$, degeneracy $g_0 = 1$) and a triplet excited state ($F = 1$, degeneracy $g_1 = 3$). Treating the hyperfine degree of freedom as a two-level system with

$$E_0 = 0, \qquad E_1 = \Delta E_{\rm hfs}^{(\rm exp)} \equiv \Delta, \tag{29}$$

the partition function is

$$Z(\beta) = g_0 e^{-\beta E_0} + g_1 e^{-\beta E_1} = 1 + 3e^{-\beta\Delta}, \qquad \beta = \frac{1}{k_B T}. \tag{30}$$

The present treatment assumes that the hyperfine manifold can be regarded as an effectively isolated internal two-level subsystem, which is appropriate for dilute atomic hydrogen under conditions where spin-exchange collisions, molecular formation, and strong external-field splittings do not dominate the equilibration.

6.2 Exact heat capacity

The mean energy is

$$\langle E \rangle = \frac{3\Delta e^{-\beta\Delta}}{1 + 3e^{-\beta\Delta}}. \tag{31}$$

Differentiation yields the exact Schottky heat capacity per atom:

$$\boxed{C_V(T) = k_B \left(\frac{\Delta}{k_B T}\right)^2 \frac{3e^{-\Delta/(k_B T)}}{\left(1 + 3e^{-\Delta/(k_B T)}\right)^2}.} \tag{32}$$

For one mole, multiply by Avogadro's number (replace k_B by R) [7, 8].

6.3 Low-temperature asymptotic (frozen regime)

For $k_BT \ll \Delta$, $e^{-\Delta/(k_BT)} \ll 1$ and (32) reduces to

$$\boxed{C_V(T) \sim 3k_B \left(\frac{\Delta}{k_BT}\right)^2 e^{-\Delta/(k_BT)} \qquad (k_BT \ll \Delta).} \tag{33}$$

In the opposite limit $k_BT \gg \Delta$, the Boltzmann weights of the singlet and triplet levels become nearly equal, the internal entropy approaches $k_B \ln(g_0 + g_1) = k_B \ln 4$, and the heat capacity again tends to zero:

$$C_V(T) \to 0 \qquad (T \to \infty). \tag{34}$$

Thus the Schottky anomaly is localized in an intermediate temperature window around $T \sim T_\Delta$.

6.4 Peak location (derived) for general degeneracy ratio

Let g_0 be the ground-state degeneracy and g_1 the excited-state degeneracy, with gap Δ. Define $r \equiv g_1/g_0$ and $x \equiv \beta\Delta = \Delta/(k_BT)$. Then

$$\frac{C_V}{k_B} = x^2 \frac{re^{-x}}{(1+re^{-x})^2}. \tag{35}$$

Let $y \equiv re^{-x}$ so that $dy/dx = -y$. Writing $(C_V/k_B) = x^2\, y/(1+y)^2$, we obtain

$$\frac{d}{dx}\left(\frac{C_V}{k_B}\right) = 2x\,\frac{y}{(1+y)^2} + x^2\,\frac{d}{dx}\left(\frac{y}{(1+y)^2}\right), \tag{36}$$

$$\frac{d}{dx}\left(\frac{y}{(1+y)^2}\right) = \left(\frac{1}{(1+y)^2} - \frac{2y}{(1+y)^3}\right)\frac{dy}{dx} = \frac{(1-y)}{(1+y)^3}\,\frac{dy}{dx}. \tag{37}$$

Using $dy/dx = -y$ gives

$$\frac{d}{dx}\left(\frac{y}{(1+y)^2}\right) = -\frac{y(1-y)}{(1+y)^3}. \tag{38}$$

Setting the derivative to zero yields the peak condition

$$\boxed{2(1+y) = x(1-y), \qquad y = re^{-x}.} \tag{39}$$

We do not claim originality for the transcendental peak equation itself; it is included here for completeness in the specific hydrogen-hyperfine context. For the physically relevant root one must have $y < 1$, since the right-hand side of (39) is positive only when $1 - y > 0$. For hydrogen hyperfine, $(g_0, g_1) = (1, 3)$ so $r = 3$. The solution of (39) was obtained by standard one-dimensional root-finding (Newton or bisection), giving

$$x_{\text{peak}} \equiv \frac{\Delta}{k_BT_{\text{peak}}} \approx 2.8449, \qquad T_{\text{peak}} \approx 0.3514\,T_\Delta \approx 24.0 \text{ mK}. \tag{40}$$

At this point

$$y_{\text{peak}} = 3e^{-x_{\text{peak}}} \approx 3e^{-2.8449} \approx 0.175 < 1,$$

so the positivity condition is self-consistently satisfied. At the peak one finds $C_V^{\max} \approx 1.02\,k_B$ per atom.

Interpretation. The hyperfine degree of freedom behaves thermodynamically as a discrete two-level subsystem: its contribution to C_V is exponentially suppressed for $T \ll T_\Delta$, peaks near $T \sim 0.35\,T_\Delta$, and vanishes again for $T \gg T_\Delta$ as the two levels become nearly equally populated.

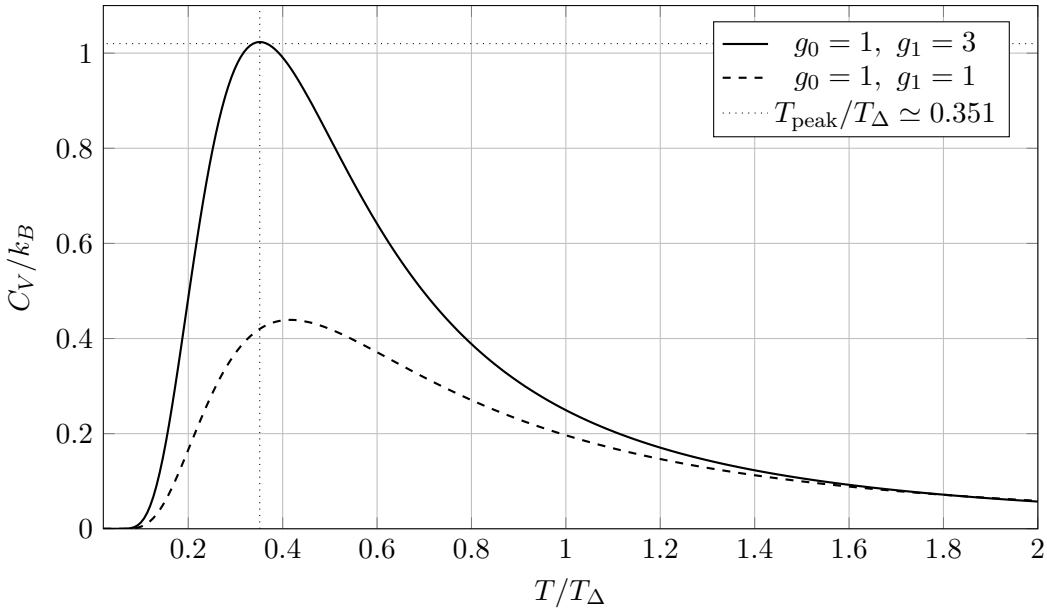

Figure 1: Dimensionless Schottky heat capacity C_V/k_B versus reduced temperature T/T_Δ for the hydrogen hyperfine doublet $(g_0, g_1) = (1, 3)$, compared with the symmetric two-level case $(g_0, g_1) = (1, 1)$. The asymmetric case peaks at $T_{\text{peak}}/T_\Delta \simeq 0.351$ with $C_V^{\max}/k_B \simeq 1.02$, whereas the symmetric case peaks at the larger temperature ratio $T/T_\Delta \simeq 0.417$.

7 Conclusion

We have presented a single self-contained derivation chain connecting the hydrogen 1s hyperfine splitting to the Schottky heat-capacity anomaly of the resulting two-level system. Starting from the dipole field with its contact term, we recovered the standard leading-order Fermi-contact gap and expressed it in closed form in terms of α, m_e, m_p, and the spin g-factors. Using the experimentally measured hyperfine energy as the activation gap, we then derived the exact canonical heat capacity for the degeneracy-asymmetric $(g_0, g_1) = (1, 3)$ system, including both low- and high-temperature limits and an explicit peak condition. The resulting millikelvin scale T_Δ and the peak temperature T_{peak} provide a direct bridge between atomic spectroscopy and low-temperature thermodynamics. Natural extensions include hydrogen-like ions, Zeeman-resolved hyperfine manifolds, and other discrete nuclear-spin systems.

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